Topic 12: Aromatic Compounds (Hydrocarbons II)

Lecture notes: what she explained and wrote on the board

CHEMISTRY (PROF. L. RANAY)

What she stressed for the quiz and NMAT

1. A multiple-choice question she explicitly flagged: **Which chemical reaction occurs first to produce meta-nitrotoluene?** Answer: **Nitration first, then alkylation** (because $-NO_2$ is a meta director, whereas $-CH_3$ is an ortho/para director). **2.** Recognize **EDG** (ortho/para directors, faster) vs. **EWG** (meta directors, slower) without a table ("not given in NMAT"): groups with a **C=O** or **multiple oxygens** (–CHO, –COR, –COOH, –COOR, $-SO_3H$, $-NO_2$) are **EWGs (meta)**; **–OH, $-NH_2$, –OR, and alkyl (–R)** are **EDGs (ortho/para)**; **halogens (–X)** are the exception (EWG in the middle of the table, but still **ortho/para**). **3.** When predicting EAS products, write **TWO products (ortho + para)** for an o/p director and **ONE product (meta)** for a meta director. **4.** Why benzene is NOT an alkene: its six π electrons are **delocalized** in a network ("no permanent address", "in unity there is strength"), so it resists addition and undergoes **electrophilic aromatic substitution (EAS)** with a **catalyst**.

BENZENE: STRUCTURE, DISCOVERY AND RESONANCE

- **Benzene (C_6H_6):** six carbons and six hydrogens (one H on each carbon of the ring); the **simplest aromatic hydrocarbon (arene)**.
  - Originally called "aromatic" because the first benzene derivatives discovered had distinct aromas: **benzaldehyde** (almonds), **vanillin** (vanilla), **cinnamaldehyde** (cinnamon). Not all aromatic compounds actually have a smell, and if another element is attached, they are called **benzene derivatives** rather than pure hydrocarbons.
- **Three scientists to keep straight:**
  - **Michael Faraday:** first **isolated** benzene from the oily residue of illuminating-gas lamps in London.
  - **August Kekulé (~50 years later):** proposed the **cyclic structure** of alternating single and double bonds after dreaming of a **snake biting its own tail** (*bitin*).
  - **Linus Pauling (1930s):** introduced the **theory of resonance**, giving the first adequate description of benzene's real structure.
- **Resonance hybrid:**
  - The two Kekulé Lewis structures with alternating single and double bonds are **contributing structures**, connected by a **double-headed resonance arrow (↔)**.
  - The real benzene molecule does **not** flip back and forth; it is a single **resonance hybrid** with all six C–C bonds **equal in length, intermediate between a single and a double bond**.
- **Delocalization ("no permanent address"):**
  - On the board she drew the six ring carbons with their **p orbitals** overlapping all around the ring into one continuous **network of π electrons**, represented by the **circle inside the hexagon**.
  - Because the double bonds are **not localized** on fixed carbons, you **cannot** number them like an alkene (no "benzene-2-ene"): every ring C–C bond is equivalent.

Note: Degrees of unsaturation. The slide says "three degrees of unsaturation" (counting only the three C=C bonds); chemically, benzene has **four** degrees of unsaturation (three π bonds + one ring). This disputed wording is kept out of the question bank.

WHY BENZENE UNDERGOES SUBSTITUTION, NOT ADDITION

- **Alkenes vs. benzene:**
  - In an **alkene**, a π bond is just a single side-by-side orbital overlap—easy to break, so alkenes readily undergo **addition** without a catalyst.
  - In **benzene**, the six π electrons form a closed delocalized network (**"in unity there is strength"**). Breaking that network by addition would destroy the stable aromatic ring.

- Therefore, benzene **resists addition** (no reaction with cold $KMnO_4$ Baeyer's reagent; no addition with Br_2 alone) and instead undergoes **substitution**, where an atom or group **replaces one of the ring hydrogens** while keeping the aromatic ring intact.
- **Every reaction of the benzene ring requires a catalyst** (such as $FeCl_3$, $FeBr_3$, H_2SO_4, or $AlCl_3$) to generate a strong electrophile and activate the reaction.

Note: Why benzene is not in the alkene family. When she asked why benzene is not classified as an alkene, she warned against answering "because it is cyclic" (cycloalkenes like cyclohexene are cyclic alkenes!). The real reason is **delocalization of its π electrons** (resonance stability), which makes it undergo **substitution instead of addition**.

NAMING AROMATIC COMPOUNDS

1. MONOSUBSTITUTED BENZENES

- **No special common name:** name the substituent as a prefix + **benzene** (one word). **No locant "1"** is written because all six carbons are equivalent:
  - **chlorobenzene**, **bromobenzene**, **nitrobenzene** ($-NO_2$ = *nitro*, not nitrogen dioxide), **ethylbenzene** ($-CH_2CH_3$).
- **Retained IUPAC parent names (memorize her list of 9):** when one of these groups is on the ring, use the special parent name instead of "benzene":
  - **toluene** ($-CH_3$, methylbenzene)
  - **styrene** ($-CH{=}CH_2$, vinylbenzene)
  - **phenol** ($-OH$, hydroxybenzene)
  - **anisole** ($-OCH_3$, methoxybenzene)
  - **aniline** ($-NH_2$, aminobenzene)
  - **benzaldehyde** ($-CHO$, aldehyde on benzene)
  - **acetophenone** ($-COCH_3$, methyl ketone on benzene)
  - **benzoic acid** ($-COOH$, carboxylic acid on benzene)
  - **benzenesulfonic acid** ($-SO_3H$, from sulfonation)
- **Phenyl group (C_6H_5- or Ph−):** when the benzene ring is treated as a **substituent (branch)** attached to a carbon chain with a higher-priority group or double bond (e.g., **1-phenylcyclohexene**, **4-phenyl-1-butene**).

2. DISUBSTITUTED BENZENES (NUMBERS AND ORTHO / META / PARA)

- **Only for TWO substituents on the ring:**
  - **1,2- = ortho- (*o*-)** (adjacent carbons)
  - **1,3- = meta- (*m*-)** (separated by one carbon)
  - **1,4- = para- (*p*-)** (opposite carbons)
- **Case A — No retained parent name:** parent is **benzene**. Give the two groups the **lowest locants**; if tied (e.g., 1,4 either way or 1,2 either way), **alphabetical order gets C1**:
  - Cl and $-CH_2CH_3$ opposite each other: **1-chloro-4-ethylbenzene** (*p*-chloroethylbenzene).
  - Br and $-NO_2$ adjacent: **1-bromo-2-nitrobenzene** (*o*-bromonitrobenzene).
  - Two identical groups: use **di-** and separate numbers with a **comma**: **1,3-dinitrobenzene** (*m*-dinitrobenzene).
- **Case B — One group gives a retained parent name:** that group is automatically **C1** (do **not** write "1" for the parent):
  - $-COOH$ at C1, $-Br$ at C2: **2-bromobenzoic acid** (*o*-bromobenzoic acid).
  - $-NH_2$ at C1, $-Cl$ at C3: **3-chloroaniline** (*m*-chloroaniline).
- **Special name for two methyls — xylene:** two $-CH_3$ groups on benzene are **never** called "di-toluene"; the retained name is **xylene** (or dimethylbenzene): **1,4-xylene** (*p*-xylene, or 1,4-dimethylbenzene).

3. POLYSUBSTITUTED BENZENES (THREE OR MORE GROUPS)

- **Never use *o*-, *m*-, or *p*- for three or more substituents**—always use numbers!
- **Rule 1:** If a retained-name group is present, it becomes the **parent name** and its ring carbon is **C1**. Number around the ring in the direction that gives the **lowest locants** at the first point of difference, then list prefixes **alphabetically**:
  - Board example A ($-CH_3$ at C1, $-NO_2$ at C2, $-Cl$ at C4): parent is **toluene** → **4-chloro-2-nitrotoluene** (chloro before nitro alphabetically).
  - Board example B ($-OH$ at C1, $-Br$ at C2, C4, C6): parent is **phenol** → **2,4,6-tribromophenol**.
  - Group table activity ($-CH_3$ at C1, $-Cl$ at C2 and C6, $-CH_2CH_3$ at C3): parent is **toluene** → **2,6-dichloro-3-ethyltoluene** (choice C).

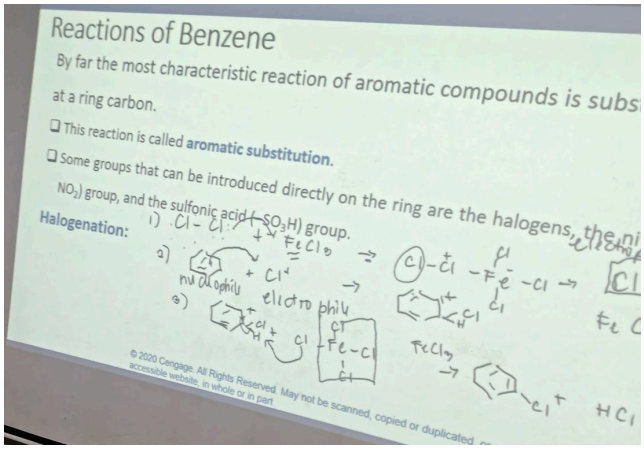
- **Rule 2:** If **no** retained parent name is present, the parent is **benzene**. Number to give the **lowest locant set** (e.g., 1,2,4 beats 1,3,4 or 1,4,5), then list substituents **alphabetically**:
  - Board example C ($-CH_2CH_3$, $-Br$, $-NO_2$ at 1,2,4): starting at ethyl gives **1,2,4** (starting at nitro gives 1,3,4) → **2-bromo-1-ethyl-4-nitrobenzene**.
- **Rule 3 — When two or more retained-name groups compete (hierarchy):** the **highest-priority functional group** wins the parent name and C1; all other groups become prefixes (**carboxylic acid > aldehyde > ketone > alcohol/phenol > amine > ether/halide/nitro**):
  - **Vanillin** has $-CHO$ (benzaldehyde), $-OH$ (phenol), and $-OCH_3$ (anisole) on the same ring. **Aldehyde beats alcohol and ether**, so the parent is **benzaldehyde** (C1 at $-CHO$): **4-hydroxy-3-methoxybenzaldehyde**.
  - **4-Aminobenzoic acid (PABA):** $-COOH$ beats $-NH_2$, so benzoic acid is the parent.

ELECTROPHILIC AROMATIC SUBSTITUTION (EAS) REACTIONS

- **Why called electrophilic aromatic substitution:**
  - **Electrophile ("electron lover"):** positively charged species (**Cl^+, Br^+, NO_2^+, SO_3H^+, R^+**) that acts as a **Lewis acid (electron-pair acceptor)**.
  - **Nucleophile ("nucleus lover"):** the **benzene ring** itself, whose π electron cloud acts as a **Lewis base (electron-pair donor)** to attack the electrophile.
  - It is an ionic **electrophilic** substitution, **not** a free-radical substitution like in alkanes.

1. HALOGENATION (AND HER 3-STEP MECHANISM ON THE BOARD)

- **Reaction:** Benzene + **Cl_2 / $FeCl_3$** → **chlorobenzene + HCl** (or **Br_2 / $FeBr_3$** → **bromobenzene + HBr**). Must use the **pure diatomic halogen (Cl_2 or Br_2)**, not HCl or HBr!
- **Her 3-step mechanism drawn on the whiteboard:**
  - **Step 1 (Make the electrophile):** Cl–Cl (Lewis base, donates e^- pair) + **$FeCl_3$** (Lewis acid, accepts e^- pair) → [Cl–Cl^+–Fe^-Cl_3] complex → **Cl^+ (chloronium ion, the electrophile)** + **$FeCl_4^-$**.
  - **Step 2 (Nucleophilic attack / EAS):** The benzene ring's π electrons (**nucleophile**) attack **Cl^+** (**electrophile**), breaking one double bond to form a **resonance-stabilized carbocation intermediate** where that ring carbon holds **both H and Cl**.
  - **Step 3 (Deprotonation & catalyst regeneration): $FeCl_4^-$** returns and takes the proton (**H^+**) to form the by-product **HCl**; the C–H bonding electrons return to the ring to **restore the aromatic benzene ring**, leaving **chlorobenzene** and regenerating the **$FeCl_3$** catalyst (not consumed).



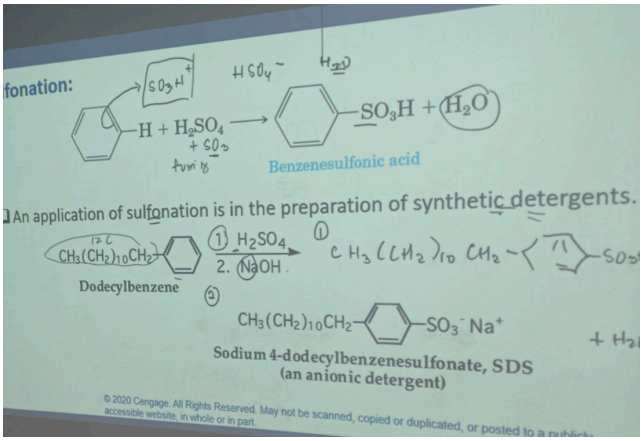
Her 3-step halogenation mechanism on the board: (1) Cl2 + FeCl3 gives Cl+ (electrophile) + FeCl4-, (2) benzene (nucleophile) attacks Cl+ to form the carbocation, (3) FeCl4- takes H+ to give chlorobenzene + HCl + FeCl3

2. NITRATION (AND NITRO REDUCTION)

- **Reaction:** Benzene + **HNO_3** (reagent) + **H_2SO_4** (catalyst) → **nitrobenzene + H_2O**.
- **Mechanism parallels halogenation:**
  - Mixing HNO_3 + H_2SO_4 produces **NO_2^+ (nitronium ion, the electrophile)** + **HSO_4^-** + **H_2O**.
  - Benzene attacks **NO_2^+** to form the carbocation holding H and NO_2; then **HSO_4^-** removes **H^+** to restore the ring and regenerate **H_2SO_4**.
- **Reduction of nitroarenes to primary arylamines:**
  - In organic chemistry, **reduction = gain of H or loss of O**; **oxidation = gain of O or loss of H**.
  - A nitro group (**$-NO_2$**) on a ring is reduced to a **primary amino group ($-NH_2$)** using **H_2 gas with a metal catalyst: Ni (nickel), Pt (platinum), or Pd (palladium)**.
  - Slide example: **4-nitrobenzoic acid + 3 H_2 (Ni/Pt/Pd) → 4-aminobenzoic acid (PABA) + 2 H_2O**. Neither the benzene ring nor the $-COOH$ group is affected—only $-NO_2$ becomes $-NH_2$!

3. SULFONATION (AND SYNTHETIC DETERGENTS)

- **Reaction:** Benzene + **concentrated H_2SO_4 + heat** (or **fuming H_2SO_4 = H_2SO_4 + SO_3 gas**, where SO_3 is the catalyst) → **benzenesulfonic acid ($C_6H_5SO_3H$) + H_2O**.
  - Electrophile she boxed on the board: **SO_3H^+**.
- **By-product rule to remember:**
  - **Halogenation and alkylation** give **HX (HCl or HBr)** as the by-product.
  - **Nitration and sulfonation** give **water (H_2O)** as the by-product.
- **Application — making synthetic anionic detergents (late 1950s, ~90% of the market):**
  - **Step 1 (Sulfonation): Dodecylbenzene** (benzene with a linear 12-carbon alkyl chain, $CH_3(CH_2)_{10}CH_2–C_6H_5$) + **$H_2SO_4$ → 4-dodecylbenzenesulfonic acid**.
  - **Step 2 (Neutralization with base):** Sulfonic acid + **NaOH → sodium 4-dodecylbenzenesulfonate (SDS)**, a water-soluble salt (anionic detergent) + **H_2O**.



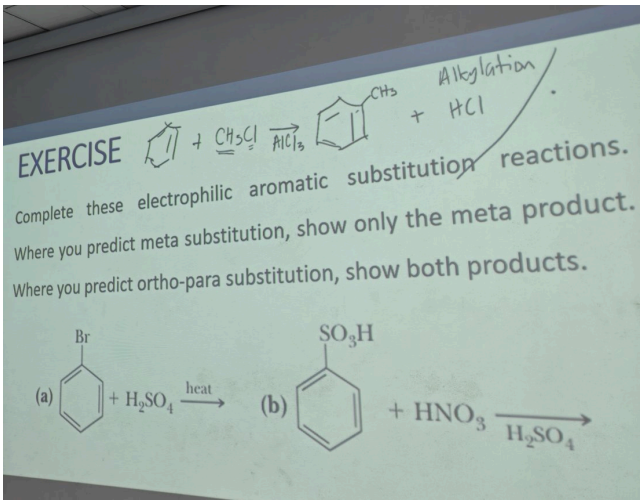
Her board notes on fuming sulfonation (SO3H+ electrophile, H2O circled) and the 2-step preparation of sodium 4-dodecylbenzenesulfonate (SDS) from dodecylbenzene

4. FRIEDEL–CRAFTS ALKYLATION (AND SIDE-CHAIN OXIDATION)

- **Alkylation reaction:** Benzene + **alkyl halide (R–Cl)** with **$AlCl_3$ (aluminum chloride) catalyst → alkylbenzene + HCl**.
  - To put a methyl group on benzene: **benzene + CH_3Cl ($AlCl_3$) → toluene + HCl**.
  - To put an ethyl group on benzene: **benzene + CH_3CH_2Cl ($AlCl_3$) → ethylbenzene + HCl**.
- **Side-chain oxidation:**
  - The benzene ring itself cannot be oxidized by $KMnO_4$ or $K_2Cr_2O_7$, **but an alkyl group attached to the ring (–CH_3, –CH_2CH_3, etc.) is oxidized all the way to a carboxyl group (–COOH)**:
  - **Toluene** (or **ethylbenzene**) + strong oxidizing agent → **benzoic acid**.

DIRECTING EFFECTS: ORTHO/PARA VS. META DIRECTORS

- When a monosubstituted benzene undergoes a second EAS reaction, **the group ALREADY on the ring** decides where the incoming electrophile goes and whether the reaction is faster or slower than plain benzene:
  - **Ortho/para directors (activate the ring → faster than benzene, except halogens):**
    - **Electron-donating groups (EDG): –NH_2, –NHR, –NR_2, –OH, –OR, –$NHCOCH_3$, –C_6H_5, –R (alkyl)**.
    - **How to spot an EDG without the table:** an **–OH, –NH, –OR**, or **alkyl carbon (–R)** bonded to the ring with **single bonds** (no C=O, no multiple oxygens).
    - **Rule for predicting products:** whenever the group on the ring is an ortho/para director (like **toluene** or **bromobenzene**), you **MUST write TWO products: the ortho (1,2) isomer AND the para (1,4) isomer**!
  - **The halogen exception (–F, –Cl, –Br, –I):**
    - Halogens sit in the **middle of the reactivity table**: they are **electron-withdrawing (slower than benzene)**, **yet they still direct to ortho and para** (write **two** products)!
  - **Meta directors (deactivate the ring → slower than benzene):**
    - **Electron-withdrawing groups (EWG): –CHO, –COR, –SO_3H, –COOH, –COOR, –CN, –NR_3^+, –NO_2** (–NO_2 is at the very bottom: slowest of all).
    - **How to spot an EWG without the table:** the atom attached to the ring has a **carbonyl double bond (C=O)** (aldehyde, ketone, carboxylic acid, ester), **two or more oxygens attached to it** (–SO_3H, –NO_2), a triple bond (–CN), or a full positive charge (–NR_3^+).
    - **Rule for predicting products:** for a meta director (like **nitrobenzene** or **benzenesulfonic acid**), write **ONLY ONE product: the meta (1,3) isomer**!



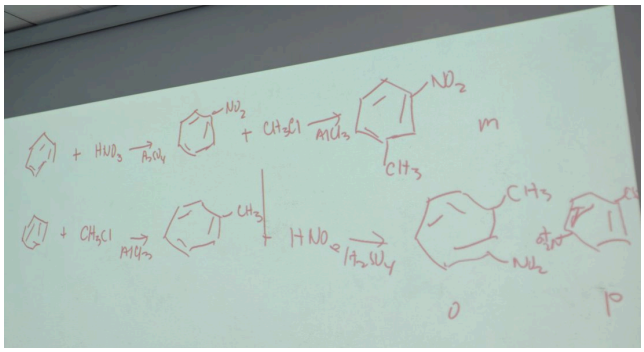
Top: her alkylation equation (Benzene + CH3Cl / AlCl3 gives toluene + HCl). Bottom: class EAS exercise — (a) bromobenzene + H2SO4/heat gives TWO products (o- and p-bromobenzenesulfonic acid); (b) benzenesulfonic acid + HNO3/H2SO4 gives ONE product (m-nitrobenzenesulfonic acid)

HER FLAGGED QUIZ ITEM: MAKING META-NITROTOLUENE

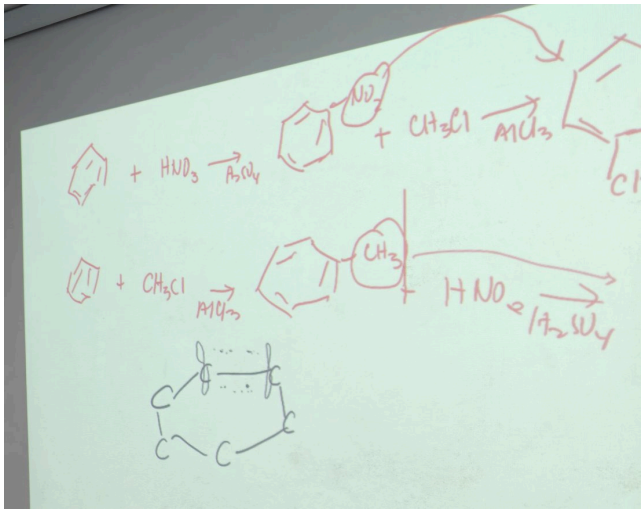
Which chemical reaction must go FIRST to synthesize meta-nitrotoluene from benzene?

Nitration first (HNO_3 / H_2SO_4), then alkylation (CH_3Cl / $AlCl_3$).

- **Route 1 — Nitration first (CORRECT for *meta*):**
  - **Step 1:** Benzene + HNO_3 / H_2SO_4 → **nitrobenzene**.
  - **Step 2:** Nitrobenzene + CH_3Cl / $AlCl_3$ → **meta-nitrotoluene** (because **$-NO_2$** is already on the ring and is an **EWG / meta director**, sending the incoming $-CH_3$ to C3).
- **Route 2 — Alkylation first (gives *ortho* and *para* instead!):**
  - **Step 1:** Benzene + CH_3Cl / $AlCl_3$ → **toluene**.
  - **Step 2:** Toluene + HNO_3 / H_2SO_4 → **ortho-nitrotoluene + para-nitrotoluene** (because **$-CH_3$** is already on the ring and is an **EDG / ortho-para director**).



Her two-route synthesis on the board: top row (nitration first, then CH3Cl/AlCl3) gives meta-nitrotoluene; bottom row (alkylation first, then HNO3/H2SO4) gives ortho- and para-nitrotoluene



Top: circled -NO2 directing CH3 to meta vs. circled -CH3 directing NO2 to ortho/para. Bottom: her sketch of the 6-carbon ring with side-by-side p orbitals forming a delocalized pi network

PHENOLS: STRUCTURE, ACIDITY, NATURAL PHENOLS AND OXIDATION

- **Structure & family:** a **hydroxyl group (–OH) bonded directly to a benzene ring carbon**.
  - **Phenols are a separate family—NOT alcohols!** (In an alcohol like benzyl alcohol, $C_6H_5CH_2OH$, the –OH is on an sp^3 carbon outside the ring.)
- **Named phenol derivatives on the slides:**
  - **Cresols** (methylphenols): e.g., **3-methylphenol = *m*-cresol**.
  - **Benzenediols (two –OH groups on benzene): 1,2-benzenediol = catechol** (*ortho*); **1,3-benzenediol = resorcinol** (*meta*); **1,4-benzenediol = hydroquinone** (*para*).
- **Acidity (the most important property of phenols):**
  - Unlike alcohols (which are neutral), **phenols are weak acids with $pK_a \approx 10$**.

- Phenols are **insoluble in water** because of the nonpolar ring, **but they react with strong bases (NaOH or KOH)** in an acid-base neutralization to form a **water-soluble phenoxide salt + water**:
  - **Phenol (C_6H_5OH) + NaOH → sodium phenoxide ($C_6H_5O^-Na^+$) + H_2O.**
- **Phenols in nature and commercial products (all tested in the slide review):**
  - **Capsaicin:** the pungent compound in **hot chili peppers**.
  - **Thymol (2-isopropyl-5-methylphenol):** antiseptic in **mouthwash and throat lozenges**; active disinfectant ingredient in **Lysol**.
  - **Vanillin (4-hydroxy-3-methoxybenzaldehyde): vanilla** flavoring extracted from vanilla pods.
  - **Urushiol:** a catechol derivative with a long 15-carbon chain; the blistering allergen in **poison ivy and poison oak** that causes skin rashes.
  - **BHA (butylated hydroxyanisole) and BHT (butylated hydroxytoluene):** synthetic **antioxidant food additives** used to prevent **spoilage** in baked goods and packaged foods.
- **Oxidation of phenols:**
  - With a strong oxidizing agent (**$K_2Cr_2O_7$ / H_2SO_4**), **phenol is oxidized to 1,4-benzoquinone (quinone)**—a diketone ring (gains O / loses H). Quinone coenzymes (like ubiquinone / CoQ) carry electrons in **cellular respiration and photosynthesis**.

SUMMARY TABLE: RETAINED AROMATIC PARENT NAMES AND EAS REACTIONS

Retained parent	Group on ring	EAS reaction	Reagents → Electrophile + By-product
Toluene	$-CH_3$ (methyl)	**Halogenation**	Cl_2 / $FeCl_3$ (or Br_2 / $FeBr_3$) → **Cl^+ (Br^+) + HCl (HBr)**
Styrene	$-CH=CH_2$ (vinyl)	**Nitration**	HNO_3 / H_2SO_4 → **NO_2^+ + H_2O** (H_2SO_4 regenerated)
Phenol	–OH (hydroxyl)	**Sulfonation**	Conc. H_2SO_4 or fuming H_2SO_4 + SO_3 → **SO_3H^+ + H_2O**
Anisole	$-OCH_3$ (methoxy)	**Alkylation**	R–Cl / $AlCl_3$ (e.g., CH_3Cl) → **R^+ + HCl**
Aniline	$-NH_2$ (amino)	**Nitro reduction**	Nitroarene + H_2 / Ni, Pt, or Pd → **primary amine ($-NH_2$)**
Benzaldehyde / Acetophenone / Benzoic acid	–CHO / $-COCH_3$ / –COOH	**Side-chain oxidation**	Alkylbenzene (toluene, ethylbenzene) + strong oxidant → **benzoic acid**

SUMMARY TABLE: DIRECTING EFFECTS IN ELECTROPHILIC AROMATIC SUBSTITUTION (NOT GIVEN IN NMAT)

Class	Groups already on the benzene ring	Directs incoming group to	Rate vs. benzene & products to write
EDG (electron-donating, activating)	**$-NH_2$, $-NHR$, $-NR_2$, –OH, –OR, $-NHCOCH_3$, $-C_6H_5$, –R (alkyl)** — single-bonded O or N, or carbon	**Ortho and para** (C2 and C4)	**Faster** than benzene · write **TWO products** (o + p)
Halogens (the exception)	**–F, –Cl, –Br, –I** — electron-withdrawing inductively, in the middle of the table	**Ortho and para** (C2 and C4)	**Slower** than benzene · write **TWO products** (o + p)
EWG (electron-withdrawing, deactivating)	**–CHO, –COR, $-SO_3H$, –COOH, –COOR, –CN, $-NR_3^+$, $-NO_2$** — C=O or atom flanked by multiple O's	**Meta** (C3)	**Slower** than benzene ($-NO_2$ slowest) · write **ONE product** (m)

Topic 13: Alcohols, Amines, Ethers and Thiols

Lecture notes: what she explained and wrote on the board

CHEMISTRY (PROF. L. RANAY)

What she stressed for the quiz and NMAT

1. Her **essay trick question**: *"Which hydrocarbon is more soluble in water?"* Answer: **Neither—ALL hydrocarbons (and thiols) are insoluble in water! 2. Predicting products on the quiz for amine + acid** (e.g., + HCl): *"Walay water!"* Copy the entire amine, **add one H to the N**, put a **+ charge on N**, and write the **anion (Cl^-)—only an ammonium salt forms, no water**. **3. NMAT favorites:** ranking boiling points (**hydrocarbons < thiols < ethers < amines < alcohols**), **Markovnikov's rule (addition)** vs. **Zaitsev's rule (elimination/dehydration)**, and the three lab oxidizing agents (**$KMnO_4$, $K_2Cr_2O_7$, PCC**) vs. the body's enzyme (**alcohol dehydrogenase**). **4. Her water-solubility carbon boundaries:** H-bonding families (**alcohols, amines**) are infinitely soluble at **C1–C3**, slightly soluble at **C4–C5**, and **insoluble at C6+**; polar non-H-bonding families (**ethers**, and later aldehydes/ketones) are **slightly soluble at C3** and **insoluble at C4+**. **5. Skipped for the quiz:** cyclic-ether IUPAC naming (only know THF and ethylene oxide by common name) and ether chemical reactions.

ALCOHOLS: STRUCTURE, CLASSIFICATION AND A-HYDROGENS

- **Definition:** an **–OH (hydroxyl) group bonded to a tetrahedral (sp^3) carbon**. Simplest = **methanol (CH_3OH)**.
  - Biological examples she gave: **glycerol (propane-1,2,3-triol)** forms the backbone of body fats and oils (**triacylglycerols / triglycerides**); **cholesterol** is an alcohol (its name ends in **-ol**!).
- **Two ways to classify alcohols as primary (1°), secondary (2°), or tertiary (3°):**
  - **By carbons attached to the C–OH carbon** (easiest for naming and physical properties):
    - **1° alcohol:** the C–OH carbon is bonded to **1** other carbon (e.g., ethanol, 1-propanol, 2,2-dimethylpropan-1-ol).
    - **2° alcohol:** the C–OH carbon is bonded to **2** other carbons (e.g., 2-propanol, 2-butanol, cyclohexanol).
    - **3° alcohol:** the C–OH carbon is bonded to **3** other carbons (e.g., 2-methyl-2-propanol).
  - **By α-hydrogens (hydrogens directly on the C–OH carbon—key for oxidation!):**
    - **1° alcohol** has **2 α-hydrogens** (can be oxidized **twice**: → aldehyde → carboxylic acid).
    - **2° alcohol** has **1 α-hydrogen** (can be oxidized **once**: → ketone, then stops).
    - **3° alcohol** has **0 α-hydrogens** (**resists oxidation**—no reaction in the lab or environment!).

NAMING ALCOHOLS (IUPAC, COMMON NAMES, AND HER BOARD DRILLS A–F)

- **IUPAC rules:**
  - **1. Parent chain:** find the **longest continuous carbon chain that CONTAINS the C–OH carbon** (even if another chain without the –OH is longer!).
  - **2. Numbering priority (her 4-step hierarchy):**
    - **(1) Nearest functional group (–OH) → (2) nearest double/triple bond → (3) nearest branch → (4) alphabetical**.
  - **3. Suffix:** drop the final **-e** of the alkane and add **-ol** (methane → **methanol**, ethane → **ethanol**). Keep **-an-** to show all C–C bonds are single (**propan-1-ol**, **2-methylpropan-2-ol**). From **C3 (propanol) upward**, always write the locator for the –OH!
  - **4. Alcohols with a C=C double bond (alkenols):** the parent chain must include **both the –OH and the C=C**, and **–OH still gets the lower number** because it is higher in the hierarchy! Drop the final ***-e*** of ***-ene*** before the vowel ***-ol***:
    - Her board example: **3-methylpent-3-en-2-ol** (numbered from the right so –OH is at C2

and C=C is at C3; written **-en-**, not *-ene-*, because *-ol* follows).

- **5. Cyclic alcohols:** when –OH is the main group on a ring, its carbon is **automatically C1—do NOT write "1"** for a single –OH on a ring (**cyclohexanol**, **2-methylcyclohexanol**).
- **6. Diols and triols: KEEP the final -e** of the parent alkane before *-diol* or *-triol* because *d* and *t* are consonants (**ethane-1,2-diol**, **propane-1,2,3-triol**, **cyclohexane-1,3-diol**).
  - Common name for a diol with two –OH groups on **adjacent** carbons: **glycol** (e.g., **ethylene glycol** = ethane-1,2-diol; **propylene glycol** = propane-1,2-diol).

- **Common names of simple alcohols (alkyl group + "alcohol"):**
  - CH_3OH = **methyl alcohol**; CH_3CH_2OH = **ethyl alcohol**; $CH_3CH_2CH_2OH$ = **propyl alcohol**; $(CH_3)_2CHOH$ = **isopropyl alcohol** (2-propanol); $CH_3CH_2CH(OH)CH_3$ = ***sec*-butyl alcohol** (2-butanol); $(CH_3)_2CHCH_2OH$ = **isobutyl alcohol** (2-methyl-1-propanol); $(CH_3)_3COH$ = ***tert*-butyl alcohol** (2-methyl-2-propanol).

HER SIX BOARD NAMING DRILLS (A–F) ANSWERED IN CLASS

Drill A: $CH_3–CH(OH)–CH_2–CH(CH_3)–CH_3$

4-methylpentan-2-ol (Table 1). Number from the end nearer –OH (C2, not C4); keep *pentan-* for single bonds.

Drill B: Cyclohexane ring with –OH and an adjacent $–CH_3$

2-methylcyclohexanol (Pantaleon's item). Never write "cyclohexan-1-ol" for a ring with one –OH—C1 is automatic!

Drill C: Seven-carbon straight chain with –OH at C2

heptan-2-ol (or **2-heptanol**; no branches).

Drill D: $(CH_3)_3C–CH_2OH$ (neopentyl alcohol skeleton)

2,2-dimethylpropan-1-ol (Table 2's debate). The longest chain containing C–OH has **3 carbons** (propan-), with two methyls on C2; you **must** write **-1-ol** because propanol has 1- and 2- isomers!

Drill E: Cyclohexane with two –OH groups at C1, C3, a methyl at C2, and an ethyl at C5

5-ethyl-2-methylcyclohexane-1,3-diol. With two –OH groups you must locate both (**-1,3-diol**), keep the **-e** of cyclohexane, and list branches **alphabetically** (*ethyl* before *methyl*).

Drill F: $CH_3–CH=C(CH_3)–C(OH)=C(CH_3)–CH_2–CH_3$

3,5-dimethylhepta-2,4-dien-4-ol. Longest chain is 7 carbons (*hepta-*); –OH is at C4 from either end (tie!), so the **second priority—nearest double bond**—decides the numbering from the left (C2,C4 instead of C3,C5).

PHYSICAL PROPERTIES: HOW TO COMPARE FAMILIES (NMAT FAVORITE)

- **Her 3-step sequence for ranking physical properties:**
  - **Step 1 — Intermolecular force (IMF): Hydrogen bonding > dipole-dipole > London dispersion**.
  - **Step 2 — Molecular size (surface area):** for the same IMF, **more carbons = larger surface area = stronger attraction = higher mp and bp**.
  - **Step 3 — For melting point ONLY: packing!** Molecules that pack more neatly into a crystal lattice (e.g., *trans* over *cis*, straight chain over branched) have a higher melting point. Say **"better packing"**, never "higher packing"!
  - **Step 3 — For boiling point: atmospheric pressure** (particles in a liquid are already far apart, so crystal packing does not apply; **boiling point is the temperature where vapor pressure = atmospheric pressure**).
- **Effect of impurities (colligative properties she reviewed):**
  - **Melting point depression:** impurities disrupt crystal packing so the solid melts at a **lower** temperature than a pure substance.

- **Boiling point elevation:** impurities create three sets of liquid interactions (solvent-solvent, solute-solvent, solute-solute) that lower vapor pressure, requiring **more heat (higher bp)** to reach atmospheric pressure.

- **Boiling-point ranking of the five families (at comparable molecular weight):**
  - **Hydrocarbons < Thiols < Ethers < Amines < Alcohols** (and once Topic 15 is included, **Carboxylic acids** top them all because of dimers!).

> **Note:** Slide typo kept out of questions. The slide table prints ethane's boiling point as `89 °C` instead of `-89 °C` (`-88.6 °C`).

- **Water solubility boundaries (polar –OH vs. growing nonpolar carbon chain):**
  - Small alcohols (**methanol, ethanol, propanol — C1 to C3**) are **infinitely soluble** ("unli-rice") in water because the H-bonding –OH overpowers the short carbon chain.
  - At **1-butanol (C4, 8 g/100 g H_2O)** and **C5**, solubility drops sharply to **slightly soluble**.
  - At **C6 and above (e.g., 1-hexanol, cyclohexanol)**, alcohols and amines are **insoluble** in water.
  - **Essay trick question she warned about:** if asked to compare the water solubility of two hydrocarbons (or thiols), **both are insoluble in water**!

REACTIONS OF ALCOHOLS

1. ACID-BASE NEUTRALITY (ALCOHOLS VS. PHENOLS)

- **Alcohols are NEUTRAL compounds** (same $pK_a \approx$ 15–18 as water; aqueous solutions have pH 7).
  - **Alcohols do NOT react with strong bases like NaOH** (unlike **phenols**, which are weak acids with $pK_a \approx 10$ and react with NaOH to form phenoxide salts!).
  - That is also why washing fishy hands with rubbing alcohol does not remove the basic amine smell—you need an acid (like vinegar/calamansi) to neutralize a base!
  - Alcohols react only with **active metals like Na** (single replacement, just like Na + H_2O) to release **H_2 gas** and form an **alkoxide ion (RO^-Na^+)**.

> **Note:** Transcript slip corrected in the bank. In class she casually said "sodium hydroxide" for alcohol + Na metal; chemically, **2 ROH + 2 Na → 2 RO^-Na^+ (sodium alkoxide) + H_2**.

2. ACID-CATALYZED DEHYDRATION (ELIMINATION → ALKENE + H_2O)

- **Reaction:** Alcohol + **conc. H_2SO_4 (or 85% H_3PO_4) + heat → alkene + H_2O**.
  - Type of reaction: **elimination** (the exact **reverse of alkene hydration**; removing water shifts the equilibrium toward the alkene).
  - **Requires at least 2 carbons** (an –OH carbon and an adjacent carbon holding an H): **methanol cannot undergo dehydration to an alkene!**
  - **Ease of dehydration: 3° alcohol (50 °C) > 2° alcohol (140 °C) > 1° alcohol (180 °C, hardest)**.
- **Zaitsev's rule (Saytzeff's rule) — which alkene is the MAJOR product:**
  - When an alcohol has two different adjacent carbons with hydrogens (like **butan-2-ol** or **2-methylbutan-2-ol**), the new C=C double bond **must form right next to where the –OH was** (never jumps carbons!).
  - **Zaitsev's rule (for ELIMINATION):** the **H is removed from the more substituted adjacent carbon** (the carbon with fewer H's / more alkyl groups), so the **major product is the more highly substituted alkene**:
    - **Butan-2-ol** + H_2SO_4/heat → **2-butene (80%, major)** + **1-butene (20%, minor)** + H_2O.
    - **2-Methylbutan-2-ol** + H_2SO_4/heat → **2-methyl-2-butene (major)** + **2-methyl-1-butene (minor)** + H_2O.
  - **Contrast with Markovnikov's rule (for ADDITION to an alkene):** in Markovnikov addition, H goes to the *less* substituted carbon; in Zaitsev elimination, H leaves the *more* substituted carbon!

3. OXIDATION OF 1°, 2° AND 3° ALCOHOLS (IN THE LAB AND IN THE BODY)

- **How to spot redox in organic chemistry:**
  - **Oxidation** = **gain of O** or **loss of H** (specifically removing an **α-hydrogen** from the C–OH carbon).
  - **Reduction** = **gain of H** or **loss of O**.
- **Products by alcohol class:**

- **1° alcohol (2 α-H):** oxidized first to an **aldehyde (–CHO)**, and oxidized further to a **carboxylic acid (–COOH)**.
- **2° alcohol (1 α-H):** oxidized to a **ketone (R–CO–R')**, which **does not oxidize further**.
- **3° alcohol (0 α-H): no reaction (resists oxidation)** because it has no α-hydrogen to remove.

- **Reagents to know:**
  - **In the laboratory (NMAT trio): $KMnO_4$** (potassium permanganate), **$K_2Cr_2O_7$** (potassium dichromate, in H_2SO_4), and **PCC** (pyridinium chlorochromate, mild oxidant that stops 1° alcohols at the aldehyde).
  - **In the human body:** the enzyme **alcohol dehydrogenase**.
- **Her three real-life oxidation pathways in the body:**
  - **1. Ethanol (beer/wine — the ONLY edible alcohol):** oxidized by alcohol dehydrogenase to **ethanal (acetaldehyde)** → oxidized further to **acetic acid (ethanoic acid = vinegar, non-toxic)**.
    - **Ethanal (acetaldehyde)** is what causes **hangover nausea, vomiting, and alcohol addiction**. Alcohol-aversion drugs (like disulfiram) block the second step (ethanal → acetic acid) so ethanal builds up right away and makes the drinker vomit!
  - **2. Isopropyl alcohol (2-propanol, rubbing alcohol):** a **2° alcohol** → oxidized in the body or by $KMnO_4$ to **acetone (propanone)** (nail-polish remover!).
  - **3. Methanol (CH_3OH, in unfiltered homemade *lambanog*):** oxidized to **methanal (formaldehyde, the component of formalin)** → oxidized further to **formic acid (methanoic acid, ant/bee venom acid)**. **Formic acid** is what destroys the optic nerve and causes **blindness** in lambanog poisoning!

AMINES: CLASSIFICATION, NAMING, BASICITY AND AMINE SALTS

- **Classification of amines (1°, 2°, 3° — watch the trap!):**
  - Unlike alcohols (where 1°/2°/3° counts carbons on the C–OH carbon), **in amines 1°, 2°, and 3° count how many CARBON groups are bonded directly to the NITROGEN atom (N)**:
    - **1° amine ($R–NH_2$): 1** carbon on N (e.g., isopropylamine, $(CH_3)_2CH–NH_2$, is a **primary amine** even though its C is 2°!).
    - **2° amine (R_2NH): 2** carbons on N.
    - **3° amine (R_3N): 3** carbons on N.
  - **Aromatic amine (arylamine):** N is bonded directly to a benzene ring (**aniline**; methylanilines are called **toluidines**; *N*-methylaniline is a 2° aromatic amine, whereas benzyldimethylamine is a 3° *aliphatic* amine because N is on a CH_2, not the ring).
  - **Heterocyclic amine:** the **N atom is part of the ring itself**:
    - *Aliphatic heterocyclic rings*: **pyrrolidine** (5-membered), **piperidine** (6-membered).
    - *Aromatic heterocyclic rings*: **pyridine**, **pyrimidine** (cytosine, uracil, thymine), **imidazole**, **purine** (adenine, guanine—the building blocks of **DNA and RNA nucleic acid bases**!).
- **Naming aliphatic amines:**
  - **IUPAC:** find the longest chain bonded to N, number nearest the C–N carbon, drop **-e**, and add **-amine** (**propan-2-amine** or 2-propanamine, **1-hexanamine**, **cyclohexanamine**).
  - **The *N*- locator for 2° and 3° amines:** any carbon group attached to the **nitrogen** (not the main chain) gets the capital letter **N-** as its locant!
    - Her board example 1: $CH_3–CH(NHCH_3)–CH_2–CH(CH_3)–CH_3$ → **N,4-dimethylpentan-2-amine** (one methyl is on N, one is on C4).
    - Her board example 2 (Chris's desk item): $CH_3–CH(CH_3)–CH(N(CH_3)CH_2CH_3)–CH_2–CH_3$ → **N-ethyl-N,2-dimethylpentan-3-amine** (C–N is at C3 from either end, so number from the side nearer the C2 methyl branch; list *N-ethyl* before *N,2-dimethyl* alphabetically).
  - **Common names:** list all alkyl groups on N **alphabetically** in one word + **amine** (e.g., **methylamine**, **isopropylamine**, **diethylmethylamine**, **triethylamine**).
- **Physical properties of amines:**
  - **Odors:** low-molecular-weight amines have **sharp, penetrating, pungent odors** (like ammonia/urine): **trimethylamine** = **rotting fish**; **putrescine** (butane-1,4-diamine) =

rotting meat; **cadaverine** (pentane-1,5-diamine) = **decaying cadavers**. Early vitamins were named **"vital amines"**.

- **Hydrogen bonding & boiling point:** 1° and 2° amines form **N–H···N hydrogen bonds** (3° amines have no N–H bond so they cannot H-bond with each other), but N is less electronegative than O (ΔEN 0.9 vs. 1.4), so **amines boil LOWER than alcohols** of similar weight, though **higher than ethers, thiols, and alkanes** (ethane -88.6 °C < methanamine -6.3 °C < methanol 65 °C).
- **Water solubility:** all amines (including 3°) H-bond with water; same teacher boundary as alcohols (**C1–C3 infinite, C4–C5 slightly soluble, C6+ insoluble**).

REACTION OF AMINES WITH ACIDS — HER FLAGGED QUIZ RULE ("WALAY WATER!")

- **Amines are BASES** because the nitrogen atom has a **lone pair of electrons** that acts as a **Lewis base (electron-pair donor)** and **Brønsted–Lowry base (proton acceptor)**.
- **Quiz rule for predicting products (Amine + Acid → Ammonium Salt ONLY, NO WATER):**
  - Copy the entire amine structure, **add one H to the nitrogen**, give the nitrogen a **+ charge**, and write the acid's **anion (e.g., Cl^- or CH_3COO^-)** beside it!
  - **1° amine ($-NH_2$) + HCl → $-NH_3^+$ Cl^-**
  - **2° amine (–NH–) + HCl → $-NH_2^+-$ Cl^-** (e.g., diethylamine + HCl → **diethylammonium chloride**)
  - **3° or heterocyclic amine (–N<) + acid → $-NH^+<$ + $anion^-$** (e.g., pyridine + acetic acid → **pyridinium acetate**)
- **Why medicines are formulated as amine salts:** large amine drugs (such as **norepinephrine / epinephrine HCl**, **diphenhydramine HCl**, **Neozep / phenylephrine HCl**) have many carbons and would be insoluble in water; converting the amine into an **ionic hydrochloride salt** makes the medicine **water-soluble** so it dissolves and travels easily in the bloodstream!

ETHERS: STRUCTURE, ALKOXY NAMING AND PHYSICAL PROPERTIES

- **Structure:** an oxygen atom bonded to **two carbon atoms (R–O–R')**. Simplest = **dimethyl ether (CH_3OCH_3)**; most common = **diethyl ether ($CH_3CH_2OCH_2CH_3$)**.
- **IUPAC naming — "Ethers are ALWAYS a branch (alkoxy)!"**
  - In IUPAC nomenclature, an ether has **no suffix** and can **never** be the parent chain!
  - **Rule:** pick the **longer carbon chain** as the **parent alkane (or alkene/ring)**, and name the shorter **–OR** group as an **alkoxy prefix** (drop -*yl*, add **-oxy**: **methoxy**, **ethoxy**, **propoxy**, **isopropoxy**, **butoxy**):
    - $CH_3CH_2–O–CH_2CH_3$ → **ethoxyethane** (common: **diethyl ether**).
    - $CH_3–O–CH_2CH_2CH_2CH_2CH_3$ → **1-methoxypentane** (common: **methyl pentyl ether**).
    - Cyclopentane ring with $–OCH_3$ → **methoxycyclopentane** (or **methoxycyclohexane** for a C6 ring; no number needed when it is the only group on the ring).
    - Her board alkene-ether example: $CH_3–CH=CH–CH(OCH_3)–CH_3$ with a methyl at C2 → **4-methoxy-2-methylpent-2-ene** (the **C=C double bond gets the lower number** C2 over the alkoxy branch at C4; alphabetically **metho-** comes before **methy-**!).

> **Note:** Slide typo kept out of questions. On the slide exercise, dicyclohexyl ether is misprinted as "cyclohexoxycyclohexane" instead of *(cyclohexyloxy)cyclohexane*.

- **Cyclic ethers (heterocyclic — O is inside the ring):**
  - She explicitly announced: **cyclic-ether IUPAC naming is NOT on the quiz!** Only recognize their common names and uses:
  - **Ethylene oxide:** 3-membered ring (2 C + O); used as a **fumigant** for foodstuffs and textiles and to **sterilize surgical instruments** in hospitals.
  - **Tetrahydrofuran (THF):** 5-membered ring (4 C + O); a common laboratory and industrial **solvent**.
- **Physical properties & reactions of ethers:**
  - Ethers have a bent C–O–C bond (weakly polar), **but NO O–H bond**, so **ether molecules CANNOT hydrogen-bond with each other**!

- Therefore, **ethers have much LOWER boiling points than isomeric alcohols** (close to alkanes of the same molecular weight: diethyl ether boils at **34.6 °C** vs. 1-butanol at **117.7 °C**).
- **Water solubility boundary (polar, no H-bond between own molecules): C3 is slightly soluble; C4 and above are insoluble**!
- **Chemical reactions of ethers: NONE for the quiz** (ethers are unreactive toward acids, bases, and oxidizing/reducing agents).

THIOLS: STRUCTURE, NAMING, STENCH AND DISULFIDE BONDS (HAIR REBONDING)

- **Structure & geometry:** contain the **–SH (sulfhydryl) group**. Because sulfur (S) is in the **same periodic group as oxygen (Group 6A)**, the C–S–H bond geometry is **bent**.
- **Physical properties:**
  - The electronegativity difference between S (2.5) and H (2.1) is only **0.4**, so the **S–H bond is non-polar** and thiols show **almost no hydrogen bonding**.
  - **Water solubility: insoluble in water** (grouped with hydrocarbons as non-polar!).
  - **Boiling point:** slightly higher than hydrocarbons (thanks to the bent shape and larger S atom), but **below ethers, amines, and alcohols** (**methanethiol 6 °C vs. methanol 65 °C**; ethanethiol 35 °C vs. ethanol 78 °C).
- **Famous stench & LPG leak detection:**
  - Low-molecular-weight thiols are notorious for **foul, rotten smells**: volcano/rotten-egg sulfur smell, **onions**, **durian**, and **skunk spray** (**2-butene-1-thiol** and **3-methyl-1-butanethiol**).
  - **Ethanethiol (CH_3CH_2SH):** listed in the *Guinness Book of World Records* as the **smelliest substance** (one drop can be smelled across a whole room). Because **LPG (liquefied petroleum gas)** is naturally **odorless**, **ethanethiol is added to LPG as a warning odorant** so you can smell a gas leak immediately!
- **Naming thiols:**
  - **IUPAC: KEEP the final -e** of the parent alkane (or alkene) and add **-thiol** (because ***t*** is a consonant): **methanethiol**, **ethanethiol**, **1-pentanethiol**, **2-methyl-1-butanethiol**, **but-2-ene-1-thiol** (the –SH carbon gets the lower number over C=C!).
  - When a higher-priority group (like –OH or –COOH) is present, –SH becomes the prefix **mercapto-** (or *sulfanyl-*): **2-mercaptoethanol** ($HO–CH_2–CH_2–SH$).
  - **Common names:** alkyl group + **mercaptan** (e.g., **ethyl mercaptan** = ethanethiol; she also mentioned *Merthiolate*).
- **Acidity and oxidation of thiols (disulfide bonds & hair rebonding):**
  - Thiols are **weak acids ($pK_a \approx 10$**, comparable to phenols) and react with strong bases like NaOH to form **thiolate salts (RS^-Na^+) + H_2O**.
  - **Oxidation of thiols → disulfides (–S–S–):** two thiol molecules (**2 R–SH**) lose 2 H (oxidation) to form a covalent **disulfide bond (R–S–S–R)**; **reduction** (+ 2[H]) breaks the –S–S– bond back into **two thiols (2 R–SH)**.
  - **Biochemistry & hair rebonding:** the amino acid **cysteine** carries an **–SH** group. Two cysteines on protein chains in **keratin** (hair and nails) cross-link via **disulfide linkages (–S–S–)**, which determine whether hair is curly or straight. **Hair rebonding** uses a **reducing agent first to break the disulfide bonds**, straightens the hair, and then uses an **oxidizing agent to re-form ("re-bond") the disulfide linkages** in their new straight positions!

SUMMARY TABLE: FIVE ORGANIC FAMILIES: INTERMOLECULAR FORCES, BOILING POINT, SOLUBILITY, AND ACID-BASE CHARACTER

Family	Strongest IMF (pure liquid)	Boiling point rank	Teacher's water solubility boundary	Acid-base character & key reaction
Alcohols (R–OH)	Hydrogen bonding (O–H, ΔEN 1.4)	**1st (highest)**	C1–C3 infinite · **C4–C5 slightly soluble** · **C6+ insoluble**	**Neutral** (like water; no rxn with NaOH; + Na metal → alkoxide + H_2)
Amines (R–NH_2, R_2NH, R_3N)	Hydrogen bonding (N–H, ΔEN 0.9; not in 3°)	**2nd**	C1–C3 infinite · **C4–C5 slightly soluble** · **C6+ insoluble**	**Basic** (N lone pair) · **+ acid → ammonium salt ONLY (no H_2O)**
Ethers (R–O–R')	Weak dipole-dipole (no O–H)	**3rd**	**C3 slightly soluble** · **C4+ insoluble**	Neutral · always an **alkoxy branch**; no reactions for the quiz
Thiols (R–SH)	Very weak dipole (bent, S–H ΔEN 0.4)	**4th**	**Insoluble** (nonpolar)	Weak acid ($pK_a \approx 10$) · **oxidation → disulfide (–S–S–)**; reduction → 2 –SH
Hydrocarbons	London dispersion only	**5th (lowest)**	**All insoluble** (essay trick question!)	Nonpolar · float on water

Topic 14: Aldehydes and Ketones

Video lecture notes: what she explained and wrote on the slides

CHEMISTRY (PROF. L. RANAY)

WHERE THIS FITS

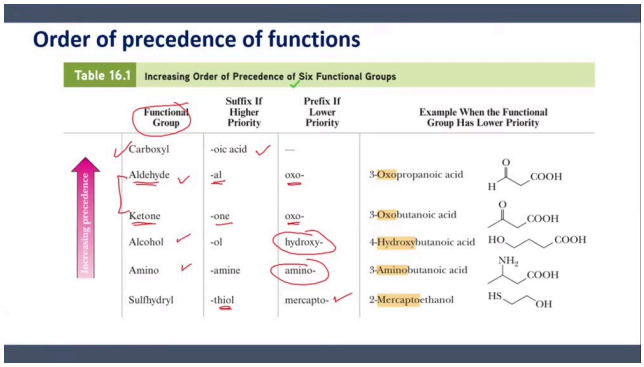
- Second-to-last family chapter: the **carbonyl** group (C=O) families. Same outline as before: structure, naming, physical properties, chemical properties.
- This chapter: aldehydes and ketones. Next chapter (Topic 15 / Unit XIV): carboxylic acids and their derivatives.

STRUCTURE

- **Aldehyde:** carbonyl carbon attached to **at least one H**. Simplest = one carbon, with H on both sides (**methanal / formaldehyde**).
- **Ketone:** carbonyl carbon attached to **two other carbons**. Smallest ketone = **three carbons: propanone (acetone)**. With only H on one side it is already an aldehyde, not a ketone.
- An aldehyde and a ketone with the same number of carbons are **functional group isomers** (same molecular formula).
  - Example she worked out: propanone and propanal are both **C_3H_6O**.
- The only difference: in an aldehyde the carbonyl carbon holds an H.

ORDER OF PRECEDENCE (NAMING PRIORITY)

- Highest to lowest: **carboxyl (-oic acid) > aldehyde (-al; prefix oxo-) > ketone (-one; prefix oxo-) > alcohol (-ol; prefix hydroxy-) > amine (-amine; prefix amino-) > thiol (-thiol; prefix mercapto-)**.
- Carboxyl is the top priority, so it is never a branch, except in very complicated structures.
- A molecule with both: aldehyde wins over ketone, so the parent name is the aldehyde.
- When OH, NH_2 or SH is present with a carbonyl, it becomes the prefix **hydroxy-**, **amino-** or **mercapto-**.

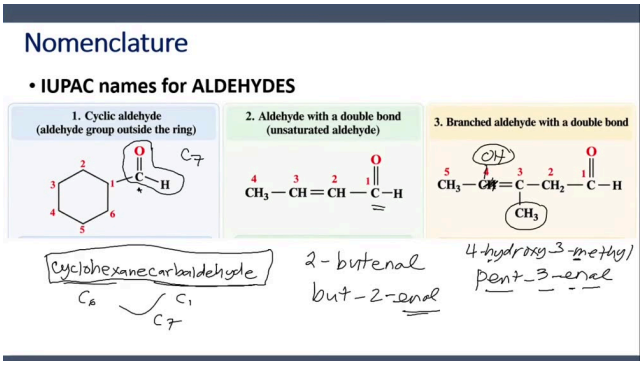


Her marks on the order-of-precedence table: circled prefixes hydroxy- and amino-, ticks down the list

NAMING ALDEHYDES

- Same rule: longest continuous chain that holds the functional group. Drop the **-e**, add **-al**.
- The CHO carbon is **always C1 and always at the end**, so **no locator** for the aldehyde.
- Her board examples (she circled the carbonyl, the OH and the numbering, and wrote the names under the structures):
  - CH_3CH_2CHO: **propanal**.
  - Branched: **3-methylbutanal**. Number from the CHO carbon.
  - With an OH lower in priority: **4-hydroxybutanal**.
- **Cyclic aldehyde: the CHO can NEVER be inside a ring.**
  - Her proof: a ring carbon already has two ring bonds; a C=O plus an H would give **five bonds**, impossible. Carbon holds only four.
  - So the CHO is always outside the ring. Name the ring, then add **carbaldehyde**: **cyclohexanecarbaldehyde**.
  - It is **C7** in total: cyclohexane is C6, the carbaldehyde carbon is C1.
- **Double bond:** the C=C stays in the parent chain, but the aldehyde still decides C1. Example: **but-2-enal** (also written 2-butenal). The "-en" and "-al" are written together.
- Her harder board example: a five-carbon aldehyde with a C=C, a methyl and an OH: **4-hydroxy-3-methylpent-3-enal**.

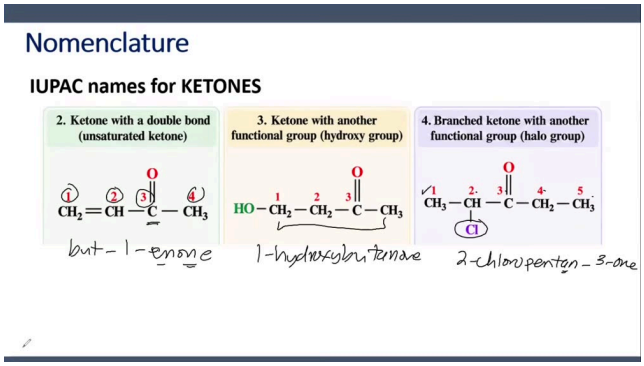
- Substituents are listed **alphabetically** (hydroxy before methyl), then the number of the double bond, then the functional group.



Her board work for aldehydes: cyclohexanecarbaldehyde (C6 + C1 = C7), but-2-enal, 4-hydroxy-3-methylpent-3-enal

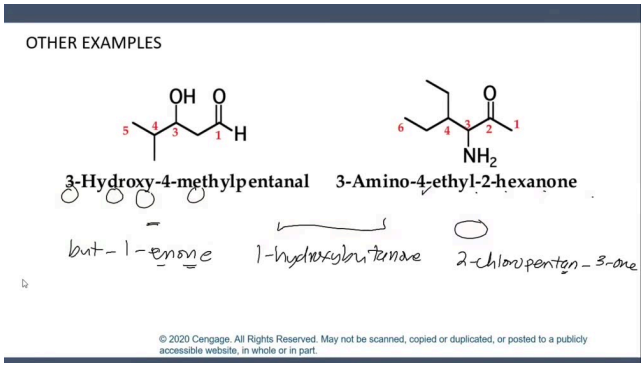
NAMING KETONES

- C=O sits **between carbons**. Drop -e, add **-one**.
- **Propanone** and **butanone** need **no locator**: there is only one possible position for the C=O (there is no 3-propanone and no 3-butanone).
- **From five carbons up you must locate it**: **pentan-2-one** and **pentan-3-one** are different compounds (older style: 2-pentanone and 3-pentanone). She showed both on the board.
- **Cyclic ketones:** the carbonyl carbon is **part of the ring**, unlike aldehydes. **Cyclohexanone**, **cyclopentanone**. No number needed: it is always carbon 1 in a ring.
- Numbering order (she repeated it): functional group first, then the double bond, then the nearest branch, then alphabetical.
  - **Highest priority is the carbonyl; the next priority is the branch.** Give the branch the lowest number you can.
- **2-chloropentan-3-one:** five carbons, so the C=O is located (3), then the chloro is on C2.
- Other examples she did: **3-hydroxy-4-methylpentanal** (aldehyde, no locator) and **3-amino-4-ethylhexan-2-one** (older style: 3-amino-4-ethyl-2-hexanone). "Hexan-2-one" is the **latest IUPAC** style.



Her board work for ketones: propanone, butanone, pentan-2-one, pentan-3-one, cyclohexanone, cyclopentanone, 2-chloropentan-3-one

Note: Her butanone shortcuts. On the board she named $CH_3–CH(CH_3)–CO–CH_3$ as "**2-methylbutanone**" (and marked "3-methylbutanone" wrong), and wrote "1-hydroxybutanone" and "but-1-enone" for butanones with an OH or a C=C. These follow her rule "no locator for butanone, so number to give the branch the lowest number". Current IUPAC writes **3-methylbutan-2-one**, **4-hydroxybutan-2-one** and **but-3-en-2-one**. The question bank does NOT quiz these disputed names. Learn her rule, but if you are asked to write a butanone with a branch, follow what she wrote in class.



Other examples: 3-hydroxy-4-methylpentanal (aldehyde circled) and 3-amino-4-ethylhexan-2-one (she wrote the newer style by hand)

PHYSICAL PROPERTIES

- Both families contain oxygen, so the **C=O is polar** (O is highly electronegative: δ− on O, δ+ on C). Both are polar molecules.
- But only **simple polar**: no H on the oxygen, so **no hydrogen bonding** between their own molecules. The intermolecular force is **dipole-dipole**.
- So they boil **lower than alcohols and carboxylic acids** (which hydrogen-bond).

- **Boiling points at similar molecular weight** (table she walked through): diethyl ether 34 < pentane 36 < butanal 76 < 2-butanone 80 < 1-butanol 117 < propanoic acid 141 °C.
  - Ether is lowest: only slightly polar.
  - Pentane (straight chain) is a little higher than the ether.
  - Butanal and butanone have a **C=O**, which is more polar than an ether, so they boil higher.
  - Butanone is slightly higher than butanal (she said some books treat them as almost equal).
  - **More branching = lower boiling and melting point** (packing).
- **Solubility:** dipole-dipole, so only some solubility. Smaller ones (formaldehyde, acetaldehyde, acetone) dissolve in water infinitely. Solubility **falls as the carbon part grows**. Comparing with ethers (same type: polar, no H bonding) she gave her boundary: **C3 slightly soluble, C4 insoluble** (confirmed in both the Whisper and the YouTube transcripts).

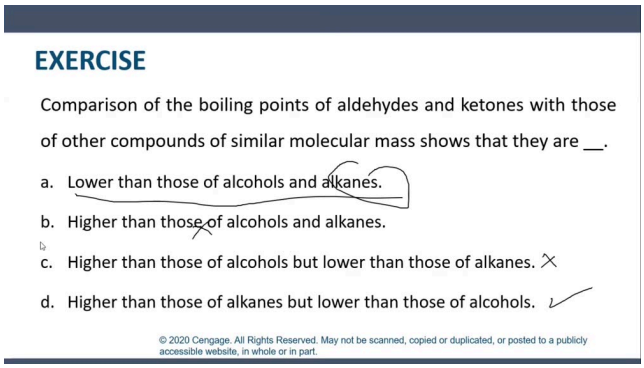
Note: The slide lists formaldehyde, acetaldehyde and acetone as infinitely soluble; her spoken boundary (C3 slightly, C4 insoluble) is the one she gave for ranking, the same as her ether numbers.

- **Exercise (answered in class): aldehydes and ketones boil higher than alkanes but lower than alcohols → letter D.**

OXIDATION AND REDUCTION (THE KEY DIFFERENCE BETWEEN THE TWO)

- **Aldehydes can be oxidized (to carboxylic acids); ketones resist oxidation.** This is how you tell them apart.
- Review she gave:
  - 1° alcohol → oxidized to **aldehyde** → oxidized again to **carboxylic acid**.
  - 2° alcohol → oxidized to **ketone** → **no further reaction**.
  - 3° alcohol → **cannot be oxidized**.
- **Reduction is the reverse** of oxidation: forward = oxidation, backward = reduction.
  - **Aldehyde → primary alcohol** (CH_2OH).
  - **Ketone → secondary alcohol** (the carbon keeps one H and two carbons).
  - Her board sketch: reduced aldehyde carbon has two H's and one carbon (primary); reduced ketone carbon has one H and two carbons (secondary).
- Oxidizing agents she named and wrote above the arrow: **$K_2Cr_2O_7$** (she said "potassium chromate"; circled on the slide), **$KMnO_4$** and **PCC**.
- She wrote **alcohol dehydrogenase** on the slide: the enzyme that does this oxidation in the body.
- Her summary on the oxidation slide, in her own shorthand:
  - **1° alcohol → [O] → aldehyde (RCHO) → [O] → RCOOH.**
  - **2° alcohol → [O] → ketone (RCOR') → [O] ✗ NR (no reaction).**
  - **3° alcohol → [O] ✗ NR.**

Note: Reason for aldehyde oxidation. She said it is "due to the acidic alpha hydrogen" that aldehydes have and ketones lack (both transcripts agree on those words). What actually matters: an aldehyde has an **H on the carbonyl carbon** (a ketone has none). The bank uses that wording.



EXERCISE

Comparison of the boiling points of aldehydes and ketones with those of other compounds of similar molecular mass shows that they are __.

a. Lower than those of alcohols and alkanes.

b. Higher than those of alcohols and alkanes.

c. Higher than those of alcohols but lower than those of alkanes.

d. Higher than those of alkanes but lower than those of alcohols.

© 2020 Cengage. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Her oxidation summary written on the slide: 1° to RCHO to RCOOH, 2° to RCOR' then no reaction, 3° no reaction; KMnO4, PCC, K2Cr2O7; alcohol dehydrogenase

TESTS THAT TELL ALDEHYDE FROM KETONE

- **Tollens' test:** ammoniacal silver nitrate ($Ag(NH_3)_2^+$). An aldehyde is oxidized and **silver deposits as a silver mirror** on the test tube. A ketone gives no mirror.
- **Benedict's test** (she also mentioned Fehling's): sugars are aldehydes, so with copper sulfate a **brick-red precipitate** shows sugar in urine. No sugar = no brick-red precipitate.

REDUCTION REAGENTS

- Lab: **H_2 with a nickel catalyst** (exercise below).
- In the body: **NADH** (nicotinamide adenine dinucleotide, reduced form). NADH is the

reducing agent and becomes NAD^+ when it oxidizes.
  - **Pyruvate (a ketone) is reduced to lactate** when the body lacks oxygen. Lactate builds up during strenuous exercise.
- Not discussed aloud: $NaBH_4$ and the hydride mechanism on the slides.

EXERCISES ANSWERED IN CLASS

What is the product when propanone is reduced with H_2 gas and a Ni catalyst?

2-propanol (a 2° alcohol). Reduction forms an alcohol; choice (a). The C=C "enols" among the choices are not alcohol products.

Which compound is resistant to oxidation with a mild oxidizing agent?

The ketone (propanone). The aldehyde and the 1° alcohol can be oxidized, so answer **(c)**.

Table 16.1 Boiling Points for Six Compounds of Comparable Molecular Weight.

Name	Structural Formula	Molecular Weight (amu)	Boiling Point (°C)
diethyl ether	$CH_3CH_2OCH_2CH_3$	74	34
pentane	$CH_3CH_2CH_2CH_2CH_3$	72	36
butanal	$CH_3CH_2CH_2CHO$	72	76
2-butanone	$CH_3CH_2COCH_3$	72	80
1-butanol	$CH_3CH_2CH_2CH_2OH$	74	117
propanoic acid	CH_3CH_2COOH	74	141

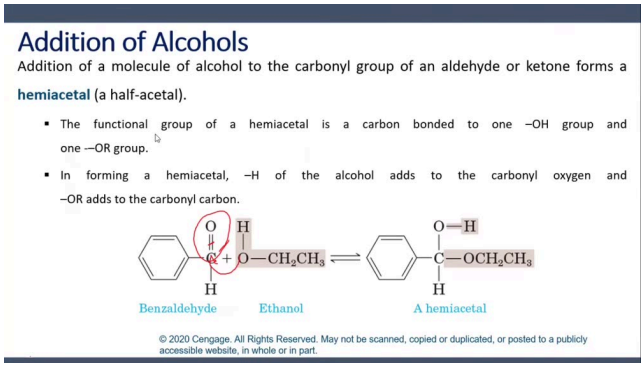
- Formaldehyde, acetaldehyde, and acetone are infinitely soluble in water.
- Aldehydes and ketones become less soluble in water as the hydrocarbon portion of the molecule increases in size.

© 2020 Cengage. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

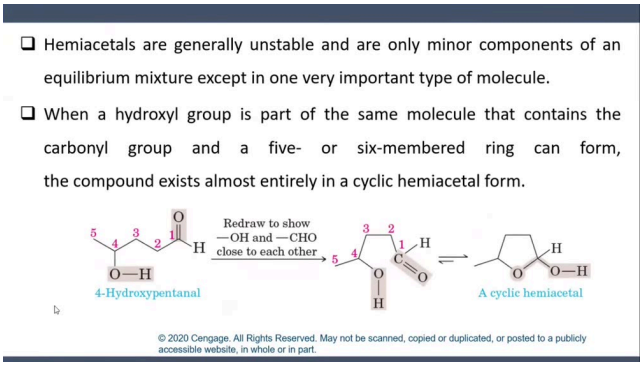
Boiling-point exercise: she marked (d) as the answer

ADDITION OF ALCOHOLS: HEMIACETALS AND ACETALS

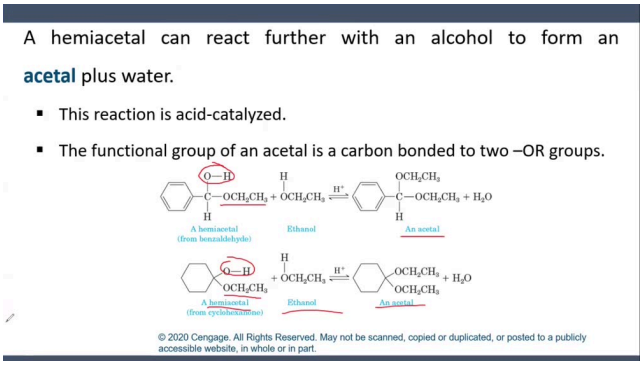
- The carbonyl has a **π bond**, so it can undergo an **addition reaction**. Here the added molecule is an **alcohol**.
- **Both aldehydes and ketones** do it (in an aldehyde the other group is H, in a ketone it is a carbon).
- **Hemiacetal:** one alcohol adds. The **H goes to the carbonyl O**; the **OR (alkoxy) goes to the carbonyl carbon**. The carbon is δ+, the oxygen of the alcohol is the electron pair donor (δ−). The π bond is gone.
  - Product: the carbon now holds **one –OH and one –OR**. "Hemi" = only half.
  - Her example: **benzaldehyde + ethanol → hemiacetal**.
- **Acetal:** a **second alcohol** adds; **the OH is replaced by OR and water is lost** (the H and the OH form H_2O). The carbon now holds **two –OR groups**.
  - Her example: the benzaldehyde hemiacetal + another ethanol → **acetal + water**.
- **What you need:** an aldehyde or ketone **plus an alcohol**.
  - If the other reagent is $KMnO_4$ (an oxidizing agent), it is **oxidation**, not hemiacetal formation.
- **Cyclic hemiacetal:** when the OH and the C=O are in the same molecule, the OH oxygen bonds to the carbonyl carbon and a ring forms.
  - Slide exercise she drew: **5-hydroxyhexanal** (OH on C5, CHO on C1). The OH oxygen bonds to C1 and closes a **six-membered ring** (5 carbons + the O). The old C=O carbon ends up holding the new **–OH and the ring O** (the hemiacetal carbon); the CH_3 stays on the ring carbon next to the O. Her red sketch showed the numbering 1 to 6 and the H leaving the OH.
  - She also drew the open-chain reaction beside it: **aldehyde + alcohol → hemiacetal**.
  - Slide: a five- (4-hydroxypentanal) or six-membered ring forms almost completely. Open-chain hemiacetals are unstable.
- Her trick for exercises: look at the C=O carbon and its oxygen. The oxygen receives the H, the O of the alcohol bonds to the carbon. If either group on the carbonyl carbon is H, it is an aldehyde-derived hemiacetal/acetal.



Her hemiacetal drawing: the carbonyl circled, H to the oxygen, OR to the carbon, and her acetal sketch with + H2O in the corner



The acetal slide with the OH and OR groups circled



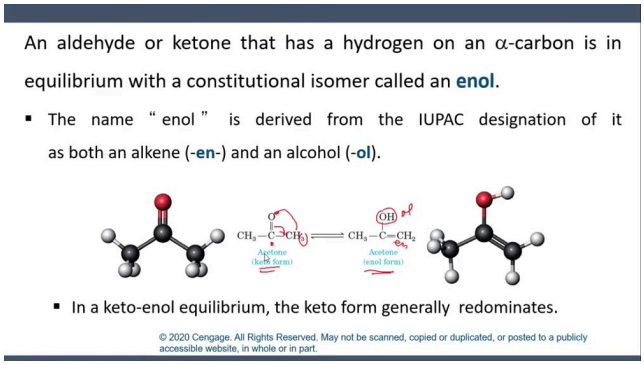
Slide exercise: 5-hydroxyhexanal closes into a six-membered cyclic hemiacetal. Her sketch: the OH oxygen bonds to the carbonyl carbon

KETO-ENOL TAUTOMERISM

- A kind of isomerism: the same substance exists as two isomers that interconvert. Left standing, it is sometimes enol, sometimes ketone.
- Needs a **C=O with an α-carbon that has an α-hydrogen**. The carbon next to the C=O is the **α-carbon**; then **β**, then **γ**.
- The **H moves to the O, the C=O π bond shifts to become a C=C**: ketone (keto form) ⇌ **enol** (alkene + alcohol, "-en" + "-ol").
- The **keto form is the more stable one**. She gave "about 70% keto, 30% enol" in class.

Note: Her 70/30 figures are not accurate for simple ketones (the keto form is far more than 70%). Only the idea "keto form predominates" is tested.

- Shows that carbonyl compounds can form **resonance-like structures**.
- **Exercise (answered in class): draw the two enol forms of each ketone.** In both (a) 2-methylcyclohexanone and (b) pentan-2-one, the new C=C can form on **either side of the C=O**, so each ketone has **two enol forms**: the C=C goes to one α-carbon or to the other.



Her two enol forms for each ketone: the C=C can form on either side of the C=O

BIOLOGICAL KETONES

- The steroid hormones **cortisol** and **aldosterone** (slide exercise): she pointed to the **ketone** in cortisol and said aldosterone has an **aldehyde together with a ketone**. Our hormones carry many carbonyl (ketone) groups.

WHAT TO REVIEW

- Order of precedence, naming (no locator for aldehyde / propanone / butanone; carbaldehyde on a ring), boiling-point ranking, oxidation vs reduction products, Tollens and Benedict tests, hemiacetal vs acetal and what is needed, keto-enol forms.

SUMMARY TABLE: ORDER OF PRECEDENCE (NAMING HIERARCHY) AND ALDEHYDE VS. KETONE REACTIONS

Priority rank	Functional group	Suffix (as parent)	Prefix (as branch) & key lecture rule
1 (highest)	**Carboxylic acid** (–COOH)	**-oic acid** (-carboxylic acid)	*carboxy-* · 'king and queen'; C1, never a branch in class items
2	**Aldehyde** (–CHO)	**-al** (ring: **-carbaldehyde**)	**oxo-** · always terminal C1 (no locator); oxidizes (Tollens silver mirror, Benedict brick-red); reduced to **1° alcohol**
3	**Ketone** (R–CO–R')	**-one**	**oxo-** · no locator for propanone/butanone; locate from **C5+**; ring ketone is C1; resists oxidation; reduced to **2° alcohol**
4	**Alcohol** (–OH)	**-ol**	**hydroxy-** · adds to aldehyde/ketone: 1 alcohol → **hemiacetal** (–OH + –OR); 2 alcohols → **acetal** (two –OR) + H_2O
5	**Amine** (–NH_2)	**-amine**	**amino-** · 1°/2°/3° counts carbons on **N**; N-branches use **N-**
6	**Thiol** (–SH)	**-thiol** (keep the -e)	**mercapto-** · foul smell (ethanethiol in LPG); cysteine –S–S– in hair rebonding

END OF LESSON